🌊 Jal-Satark: Regional Water & Health Intelligence Platform

Smart Community Health Surveillance, Water Quality Early Warning & Emergency Response System

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🛠️ 1. Tools & Technology Stack

Layer	Technologies & Libraries	Purpose
Frontend	**React 18, Vite, Tailwind CSS**	High-performance Single Page Application with dynamic role-based theme styling (Accessible clean mode for citizens / Ultra-dark tactical command palette for District Admins).
Mapping & Spatial GIS	**Leaflet, React-Leaflet, CartoDB Tiles**	Interactive Northeast regional map with dynamic basemap switching (CartoDB Dark Matter for Admin / CartoDB Voyager for Citizens), 62 live monitoring stations, geofence rings, and state jump controls.
Client-Side Cryptography	**Web Crypto API** `window.crypto.subtle`	Zero-knowledge End-to-End Encryption (E2EE) utilizing **RSA-OAEP (2048-bit)** key wrapping and **AES-GCM (256-bit)** authenticated symmetric ciphering for citizen whistleblower dispatches.
Speech & Vision UI	**Web Speech API, HTML5 Geolocation API, FileReader/Canvas**	Multi-dialect voice-to-text recognition (English, Hindi, Bengali, Assamese), live GPS tracking radar, and optical water sample image upload.
Backend API	**Node.js, Express.js (ES Modules)**	Modular RESTful API backend handling telemetry ingestion, ASHA surveillance, RBAC authentication, E2EE key exchange, and audit queries.

Real-Time Streaming	**Socket.io (WebSockets)**	Bi-directional event broadcasting for real-time telemetry updates, emergency siren broadcasts, and instant outbreak cluster alerts.
Authentication & RBAC	**JWT (JSON Web Tokens), Bcryptjs**	Secure password hashing (10 salt rounds) and role-gated middleware enforcing authorization across `District Health Admin`, `ASHA Worker`, and `Citizen` roles.
Database & GIS	**PostgreSQL + PostGIS** *(with automatic zero-dependency in-memory fallback store `dbStore.js`)*	Geospatial queries (`ST_DWithin`, geometry points), historical sensor telemetry, user public keys, and continuous learning consultation records.
ML & Predictive Engine	**Python, FastAPI, Scikit-Learn, Pandas, NumPy, Joblib**	Dual Random Forest classifiers for outbreak risk scoring and contamination prediction trained on 20,000 multi-modal records (`jal_shield_synthetic_20000.csv`).
Local GenAI Chatbot	**Ollama (Llama-3)** *(with intelligent deterministic clinical fallback engine)*	Edge-deployable clinical triage assistant providing hyper-local diagnosis across English, Bengali, Hindi, and Assamese.
Computer Vision (Simulated)	**Pillow (PIL)**	Optical clarity analyzer calculating RGB histograms, estimated turbidity (NTU), algal discoloration, and clinical dehydration flushing.

✨ 2. Comprehensive Implemented Features

📡 1. Dynamic Northeast Multi-State Monitoring Grid (62 Stations)

Real-World Dataset Integration: Automatically parses 62 distinct monitoring stations across **Meghalaya (32), Assam (15), Manipur (6), Mizoram (6), and Nagaland (3)** from the 20,000-record dataset.
Panoramic Regional Map Framing: Default view centers at `[25.57, 92.50]` (Zoom: `7.5`), framing all five Northeast states simultaneously.
State & Regional Jump Controls: Smooth 1-click animated panning (`map.flyTo`) to state clusters (Assam Basin, Meghalaya Hills, Manipur Valley, Mizoram Ridge, Nagaland Heights).
Dynamic Risk-Level Color Coding: Markers styled by live risk rating:
🟢 **Emerald Green (`#10B981`)**: Low Risk / Safe Potable Water.
🟡 **Amber (`#F59E0B`)**: Moderate Risk / Precautionary Alert.
🔴 **Emergency Red (`#EF4444`)**: High Contamination / Outbreak Cluster with animated pulsing ring and 1.5 km geofence radius.

👷 2. ASHA Field Surveillance & Citizen Proxy Registration Desk

Dedicated Field Desk: Role-gated portal designed specifically for Accredited Social Health Activists (ASHAs) and grassroots community health workers.
Household Proxy Intake: Log patient symptoms on behalf of rural or ward families who lack smartphones or internet connectivity.
Vulnerable Group Tracking: Demographics capture (*Children under 5, Elderly 60+, Pregnant/Nursing, Pre-existing conditions*).
Field Action & Medical Logistics: Record administered interventions (*Distributed ORS packets, Dispatched Halazone Chlorine disinfection tablets, Referred to PHC, Rapid Rehydration*).
⚡ **Emergency Cluster Flagging**: Single-tap toggle that instantly triggers high-priority incident triage alerts across District Administrator command consoles.

🗄️ 3. Master Database Explorer & Unified Audit Console

Full Database Observability: High-density data console for District Health Admins providing live querying across all 7 core partitions:

- 💧 `telemetry` (`telemetry_readings`)
- 📋 `symptoms` (`symptom_reports`)
- 🤖 `consultations` (`chat_consultations`)
- 🔒 `encrypted_reports` (`encrypted_citizen_reports`)
- 📢 `advisories` (`advisories`)
- 🚨 `alerts` (`alerts`)
- 👥 `users` (`users`)

Real-Time Search & Filtering: Multi-column live search, district filters, severity filters, and pagination.

Interactive Record Inspector Modal: Click any record to launch an in-depth modal featuring a structured property view and syntax-highlighted raw JSON with 1-click clipboard copy.

RFC-4180 CSV & JSON Exports: 1-click download of filtered database records for offline epidemiological research.

🔒 4. End-to-End Encrypted (E2EE) Whistleblower & Citizen Dispatch

Client-Side Cryptography: Zero-knowledge encryption running directly in the citizen's browser using native Web Crypto API.

Hybrid Encryption: Combines **RSA-OAEP (2048-bit)** key wrapping with **AES-GCM (256-bit)** payload encryption.

Official Classified Inbox: District Health Admins can view public key fingerprints, securely decrypt payloads in volatile memory, and update investigation statuses.

Anonymous Reporting Mode: Allows whistleblowers to report illegal industrial effluent dumping or municipal pipeline tampering without exposing their identity.

🎨 5. Dynamic Role-Based Theme Styling

District Health Admin High-Command Aesthetic: Automatically applies deep tactical obsidian background (`#070B14`), translucent slate-navy HUD surfaces, neon teal borders, and CartoDB Dark Matter map tiles.

Citizen / Public Mode: Clean, accessible white-and-navy emergency health aesthetic with CartoDB Voyager map tiles.

Live Status Indicator Bars: Real-time permission pills displayed at the very top of the navigation bar.

🩺 6. Jal-Swasthya AI: Vernacular Health Assistant & Triage

Local Llama-3 LLM: Runs private, zero-cloud health triage directly on-premises via local Ollama.
Vernacular Support: Multi-lingual clinical triage supporting **English, Bengali, Hindi, and Assamese**.
Voice-to-Text Speech Recognition: Built-in microphone with real-time transcription synchronized with the selected language dialect.
Computer Vision Optical Analysis: Upload water sample photos or skin cues; the vision engine assesses turbidity, algal suspension, and clinical flushing.
Hyper-Local Contextualization: Combines GPS coordinates with nearest-station water telemetry so the AI diagnosis factors in live water quality.
Continuous Learning Consultation Registry: Archives triage cases with 1-click verification for live Random Forest model retraining.

🚨 7. Transparent, Deterministic Early Warning Alerts

Standardized WHO/BIS Thresholds: Transparent alert logic without black-box metrics (Turbidity $>5.0\text{ NTU}$, pH out of range <6.5 or >8.5, TDS $>500\text{ ppm}$).
Spatial Symptom Clustering: Automatic cluster detection triggering when >5 gastrointestinal cases occur within a 2 km radius of a water point.
Human-Readable Alert Cards: Plain-language explanations detailing exact root-cause metrics and immediate corrective action checklists.
Audio-Visual Siren: Integrated web synthesizer generating priority emergency warning tones for critical hazards.

🛡️ 3. How Data is Kept Safe from Malicious Entities

Data security, citizen privacy, and institutional integrity are foundational pillars of the Jal-Satark platform:


[Citizen Browser]
[Jal-Satark Backend]
[District Admin Console]
|--- 1. Fetch Official RSA Public Key ------>|
|<-- 2. Return Admin Public Key -------------|
[Local Client-Side Encryption]
- Generate 256-bit AES-GCM Key
- Encrypt Payload with AES-GCM
- Encrypt AES Key with RSA-OAEP
|--- 3. Dispatch Encrypted Ciphertext ------>|
(Zero Plaintext Sent to Server)
(Stores Ciphertext Only)
|<--- 4. Query Encrypted Reports ---------|
|--- 5. Deliver Encrypted Ciphertext ---->|
[Local Admin Decryption]
- Decrypt AES Key with Private Key
- Decrypt Payload in Volatile RAM
- Plaintext NEVER Leaves Memory

1. Zero-Knowledge End-to-End Encryption (E2EE)

Client-Side Cryptography: Whistleblower reports and sensitive hazard tips are encrypted exclusively on the sender's client device using `window.crypto.subtle`.
Hybrid Cryptographic Standard: Uses 2048-bit RSA-OAEP (SHA-256) for asymmetric key wrapping and 256-bit AES-GCM for payload encryption.
Zero-Knowledge Backend Storage: The backend server and database only ever receive, store, and transmit ciphertext strings and initialization vectors (IVs). Even if an attacker compromises the server or database, the reports cannot be decrypted without the Admin's private key.
Volatile Memory Plaintext Handling: Plaintext data is decrypted solely within the District Admin's browser session and is never persisted in browser storage or sent back to the server.

2. Strict Role-Based Access Control (RBAC) & Middleware Guards

Cryptographic JWT Tokens: User authentication utilizes JSON Web Tokens signed with HMAC-SHA256 and verified on every sensitive route.
Granular Privilege Gating:
`District Health Admin`: Gated access to broadcast advisories, Master Database Explorer, classified inbox, and model retraining.
`ASHA Worker`: Gated access to household proxy intake and field medical log submission.
`Citizen`: Gated access to public telemetry, E2EE dispatch, and health AI triage.
403 Forbidden Middleware: Any unauthorized attempt to call privileged endpoints (e.g. citizen calling `/api/advisories` or `/api/admin/db/overview`) is rejected immediately at the API gateway layer.

3. Credential Security & Audit Log Sanitization

Bcrypt Password Hashing: User passwords are encrypted with `bcryptjs` using a salt work factor of 10. Passwords are never stored in plaintext.
Automated Memory Sanitization: All admin database queries, API responses, and CSV/JSON export routines strictly sanitize user records, permanently stripping `password_hash` and `plain_password_fallback`.
Pseudonymous Patient Identifiers: Community symptom logs and clinical records use pseudonymous hashes (`PT-1049`, `HH-BORA-128`) rather than PII (Personally Identifiable Information), adhering to HIPAA and healthcare data privacy best practices.

4. Edge & On-Device AI Privacy Isolation

No Third-Party AI Data Leakage: All ML inference (Scikit-Learn Random Forest) and GenAI triage (local Llama-3 / fallback engine) execute strictly on-premises or within the local network perimeter.
Zero External API Transmission: Zero patient symptom descriptions, location pings, or water quality telemetry are ever sent to commercial public AI APIs (such as OpenAI, Anthropic, or external cloud providers).

5. Injection Hardening & Secure Querying

Parameterized SQL Queries: All database queries through PostgreSQL use parameterized bindings (`$1, $2, ...`), neutralizing SQL Injection (SQLi) vectors.
RFC-4180 CSV Sanitization: Export routines escape and quote all values, neutralizing CSV formula injection attacks.
Input Validation: Strict schema validation ensures that malformed payloads, out-of-range sensor readings, or invalid geographic coordinates are rejected before processing.

🚀 4. Future Roadmap & Scalability

🪜 Scalability & Infrastructure Expansion

1. **IoT Edge Gateway Integration**:

   Deploy lightweight MQTT / CoAP brokers on low-power LoRaWAN and GSM edge gateways for remote rural sensor nodes with solar telemetry.

1. **Distributed Cloud Architecture**:

   Containerize microservices with Docker and orchestrate on Kubernetes (K8s) with Apache Kafka for high-throughput streaming across thousands of district nodes.

1. **Federated Learning on Edge**:

   Enable decentralized model training on local PHC (Primary Health Centre) servers to improve diagnostic accuracy without transmitting sensitive patient data.

🔮 New High-Impact Features

1. **Satellite & Meteorological Ingestion**:

   Ingest live Sentinel-2 multispectral imagery and meteorological rainfall/flood forecasts to predict runoff-induced contamination 48 hours in advance.

1. **Automated Citizen Broadcast via SMS & IVR**:

Integrate cellular SMS gateway and automated Interactive Voice Response (IVR) voice calls in local dialects to warn non-smartphone users during boil-water alerts.

1. **Pathogen DNA / Microfluidic Sensor Hookup**:

   Interface with emerging microfluidic PCR water sensors for rapid detection of specific pathogens (*Vibrio cholerae*, *Salmonella enterica*, *Rotavirus*).

1. **Supply Chain & Chlorine Logistics Dispatcher**:

   Automated procurement and dispatch tracker for water purification tablets (Halazone), ORS kits, and mobile chlorination tankers to affected outbreak zones.